

Regression Modeling and Inference

STA 427/527, Fall 2019, Xin Wang

4.1 Bayesian linear regression

4.1.1 Review of linear regression in frequentist perspective

$$\begin{aligned} Y &= f(X_1, X_2, \dots, X_k) + \epsilon \\ &= \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon \end{aligned}$$

Suppose we have n observations, $(Y_i, X_{i1}, X_{i2}, \dots, X_{ik})$ is the i th observation, for $i = 1, \dots, n$. The corresponding model is

$$Y_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_k X_{ik} + \epsilon_i, i = 1, \dots, n$$

where $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$

In a matrix approach, $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon} \sim MVN(0, \sigma^2 \mathbf{I})$, where

$$\mathbf{I}_{n \times n} = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}$$

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}_{n \times 1}, \quad \mathbf{X} = \begin{pmatrix} 1 & X_{11} & \dots & X_{1k} \\ 1 & X_{21} & \dots & X_{2k} \\ \vdots & \vdots & & \vdots \\ 1 & X_{n1} & \dots & X_{nk} \end{pmatrix}_{n \times (k+1)}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}_{(k+1) \times 1}, \quad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}_{n \times 1}$$

- Distribution of $\boldsymbol{\epsilon}$

$$\begin{aligned} f(\boldsymbol{\epsilon} | \sigma^2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n \epsilon_i^2\right\} \\ \downarrow &= (2\pi)^{-n/2} |\sigma^2 \mathbf{I}|^{-1/2} \exp\left\{-\frac{1}{2} \boldsymbol{\epsilon}^T (\sigma^2 \mathbf{I})^{-1} \boldsymbol{\epsilon}\right\} = (2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2} \boldsymbol{\epsilon}^T \boldsymbol{\epsilon}\right\} \\ &\text{matrix form} \end{aligned}$$

- Conditional distribution of $\mathbf{Y}$? $\mathbf{Y} | \mathbf{X}, \boldsymbol{\beta}, \sigma^2 \sim MVN(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$

- Data distribution: $\rightarrow$ this is the distribution that we are interested in

$$f(\mathbf{y} | \boldsymbol{\beta}, \sigma^2) = \frac{1}{\sqrt{(2\pi)^n (\sigma^2)^n}} \exp\left[-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right]$$

$$\begin{aligned} \rightarrow iid \quad f(\mathbf{y} | \boldsymbol{\beta}, \sigma^2) &= \prod_{i=1}^n f(y_i | \boldsymbol{\beta}, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2\right\} \\ \downarrow &= (2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right\} \\ &\text{matrix form} \end{aligned}$$

- Solution of $\boldsymbol{\beta}$, $\mathbf{B} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$

$$\rightarrow \text{predicted value} \rightarrow \hat{y}_i = \mathbf{x}_i^T \mathbf{B}$$

$$\begin{aligned} \rightarrow \text{least squares} \rightarrow SSE &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \mathbf{x}_i^T \mathbf{B})^2 \\ \downarrow &= (\mathbf{y} - \mathbf{X}\mathbf{B})^T (\mathbf{y} - \mathbf{X}\mathbf{B}) \\ &\text{matrix form} \end{aligned}$$

$$\rightarrow \text{Goal} \rightarrow \text{Find } \mathbf{B} \text{ that can minimize } (\mathbf{y} - \mathbf{X}\mathbf{B})^T (\mathbf{y} - \mathbf{X}\mathbf{B})$$

- $SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = (y - XB)^T (y - XB)$

↓

$$= y^T y - 2B^T x^T y + B^T x^T x B \quad \text{c "fail" matrix form?}$$

$$\rightarrow \frac{\partial SSE}{\partial B} = -2x^T y + 2x^T x B$$

$$\rightarrow 0 = \dots$$

$$\Rightarrow x^T x B = x^T y \Rightarrow B = (x^T x)^{-1} x^T y$$

↳ That is our formula of estimate of β

$$\rightarrow \text{Residuals} \rightarrow e_i = y_i - \hat{y}_i, \quad e = [e_1, e_2, \dots, e_n]^T$$

$$\text{matrix form} \rightarrow e = y - XB$$

$$= y - x(x^T x)^{-1} x^T y$$

$$= (I - H) y, \quad \text{where hat matrix } H = x(x^T x)^{-1} x^T$$

$$\rightarrow \text{properties} \rightarrow \text{Idempotent} \Rightarrow H^2 = H$$

$$\hookrightarrow \text{proof} \Rightarrow H \cdot H = [x(x^T x)^{-1} x^T] [x(x^T x)^{-1} x^T]$$

$$\downarrow$$

$$= x(x^T x)^{-1} x^T$$

$$\Rightarrow (I - H)^2 = I - H$$

$$\rightarrow SSE = e^T e$$

$$= [(I - H)y]^T [(I - H)y]$$

$$= y^T (I - H)^T y$$

$$= y^T (I - H) y$$

$$\hat{\sigma}^2 = \frac{SSE}{df(SSE)}$$

$$\downarrow$$

$$= \frac{SSE}{n - (k+1)} \rightarrow \text{That is our estimate of } \sigma^2$$

Now about our estimate of β

$$\rightarrow V(B) = V[(x^T x)^{-1} x^T y]$$

$$= (x^T x)^{-1} x^T V(y) x (x^T x)^{-1}$$

↳ x 's are known constants, y 's have variation

$$+ V(ax) = a^T V(x)$$

⇒ just using matrix form

$$V(ax) = a V(x) a^T$$

↳ move out front for convenience

$$= \sigma^2 I (x^T x)^{-1} x^T x (x^T x)^{-1}$$

↳ inverse

$$= \sigma^2 I (x^T x)^{-1}$$

↳ drop I

$$= \sigma^2 (x^T x)^{-1}$$

4.1.2 The standard noninformative prior distribution

- $\pi(\beta, \sigma^2) \propto (\sigma^2)^{-1}$

$$\rightarrow \text{Derivation} \rightarrow \pi(\beta, \sigma^2) = \pi(\beta) \pi(\sigma^2) \quad \text{c. ll priors}$$

$$\int \pi(\beta) d\beta = 1$$

$$\pi(\beta, \sigma^2) \propto 1$$

$$\Rightarrow \pi(\beta) \propto (\sigma^2)^{-1}$$

See multiparameter models, Math, page 3

- Data distribution:

$$\rightarrow f(y | \beta, \sigma^2) \propto (\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} (y - x\beta)^T (y - x\beta) \right\}$$

(matrices)

↳ rewrite this

$$\rightarrow \dots = \underbrace{(y - x\beta + x\beta - x\beta)^T}_{a} \underbrace{(y - x\beta + x\beta - x\beta)}_{b}$$

↳ add in parts = 0 & pay attention to β vs $\hat{\beta}$

$$\Rightarrow \text{call } B = (x^T x)^{-1} x^T y$$

$$= (\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} [SSE + (B - \beta)^T x^T x (B - \beta)] \right\}$$

Beneficial rewriting b/c now we have the likelihood in terms of β which makes next posterior step easier!

Data part
(β vs $\hat{\beta}$)

Parameter-dependent part
* note: equivalent to having $(B - \beta)$ w/ all squares Δ^2

$$= \underbrace{(y - x\beta)^T (y - x\beta)}_{SSE} + \underbrace{2(y - x\beta)^T x (B - \beta)}_{\text{factor out } x} + \underbrace{(B - \beta)^T x^T x (B - \beta)}_{\text{factor out } x}$$

$$\hookrightarrow (y - x\beta)^T x = (y - x(x^T x)^{-1} x^T y)^T x$$

$$= y^T [I - x(x^T x)^{-1} x^T]^T x$$

$$= y^T [x - x(x^T x)^{-1} x^T x]$$

$$= y^T [x - x]$$

$$= 0$$

↳ distribute x on RHS into $[\]$

- Joint posterior: $p(\beta, \sigma^2 | y) \propto$ data dat * prior dat

$$= (\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} [SS\bar{E} + (\beta - B)^T X^T X (\beta - B)] \right\} * (\sigma^2)^{-1}$$

$$= (\sigma^2)^{-n/2-1} \exp \left\{ -\frac{1}{2\sigma^2} [SS\bar{E} + (\beta - B)^T X^T X (\beta - B)] \right\}$$

- The conditional posterior of $\beta | \sigma^2, y \rightarrow p(\beta | \sigma^2, y) =$ joint posterior / marginal posterior

$$= \frac{p(\beta, \sigma^2 | y)}{p(\sigma^2 | y)}$$

$$= \frac{p(\beta, \sigma^2 | y)}{(\sigma^2)^{-n/2-1} \exp \left\{ -\frac{1}{2\sigma^2} [SS\bar{E} + (\beta - B)^T X^T X (\beta - B)] \right\}}$$

$$\propto \exp \left\{ -\frac{1}{2\sigma^2} (\beta - B)^T X^T X (\beta - B) \right\}$$

$$\propto \exp \left\{ -\frac{1}{2} (\beta - B)^T [\sigma^2 (X^T X)^{-1}]^{-1} (\beta - B) \right\}$$

$$\sim \text{MVN}(B, \sigma^2 (X^T X)^{-1})$$

$$\Rightarrow \text{let } z \sim \text{MVN}(\mu, \Sigma) \Rightarrow f(z) = (2\pi)^{-n/2} |\Sigma|^{-1/2} \exp \left\{ -\frac{1}{2} (z - \mu)^T \Sigma^{-1} (z - \mu) \right\}$$

$$\downarrow \Rightarrow \exp \left\{ -\frac{1}{2} (z - \mu)^T \Sigma^{-1} (z - \mu) \right\}$$

- The marginal posterior of σ^2 given y

$$\rightarrow p(\sigma^2 | y) = \int p(\beta, \sigma^2 | y) d\beta$$

$$= \int (\sigma^2)^{-n/2-1} \exp \left\{ -\frac{1}{2\sigma^2} [SS\bar{E} + (\beta - B)^T X^T X (\beta - B)] \right\} d\beta$$

$$= (\sigma^2)^{-n/2-1} \exp \left\{ -\frac{1}{2\sigma^2} SS\bar{E} \right\} \int \exp \left\{ -\frac{1}{2\sigma^2} (\beta - B)^T X^T X (\beta - B) \right\} d\beta$$

$$\propto (\sigma^2)^{-n/2-1} \exp \left\{ -\frac{1}{2\sigma^2} SS\bar{E} \right\} \int \exp \left\{ -\frac{1}{2\sigma^2} (\beta - B)^T X^T X (\beta - B) \right\} d\beta$$

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$$\propto (\sigma^2)^{-n/2-1} \exp \left\{ -\frac{1}{2\sigma^2} SS\bar{E} \right\} \int \exp \left\{ -\frac{1}{2\sigma^2} (\beta - B)^T X^T X (\beta - B) \right\} d\beta$$

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$$\propto (\sigma^2)^{-n/2-1} \exp \left\{ -\frac{1}{2\sigma^2} SS\bar{E} \right\} \int \exp \left\{ -\frac{1}{2\sigma^2} (\beta - B)^T X^T X (\beta - B) \right\} d\beta$$

$$\propto (\sigma^2)^{-n/2-1} \exp \left\{ -\frac{1}{2\sigma^2} SS\bar{E} \right\} \int \exp \left\{ -\frac{1}{2\sigma^2} (\beta - B)^T X^T X (\beta - B) \right\} d\beta$$

$$\propto (\sigma^2)^{-n/2-1} \exp \left\{ -\frac{1}{2\sigma^2} SS\bar{E} \right\} \int \exp \left\{ -\frac{1}{2\sigma^2} (\beta - B)^T X^T X (\beta - B) \right\} d\beta$$

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$$\propto (\sigma^2)^{-n/2-1} \exp \left\{ -\frac{1}{2\sigma^2} SS\bar{E} \right\} \int \exp \left\{ -\frac{1}{2\sigma^2} (\beta - B)^T X^T X (\beta - B) \right\} d\beta$$

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$$\propto (\sigma^2)^{-n/2-1} \exp \left\{ -\frac{1}{2\sigma^2} SS\bar{E} \right\} \int \exp \left\{ -\frac{1}{2\sigma^2} (\beta - B)^T X^T X (\beta - B) \right\} d\beta$$

- Sampling from the posterior distribution: Two steps

$$1. \text{ Step 0 } \rightarrow B = (X^T X)^{-1} X^T y$$

$$SS\bar{E} = (y - XB)^T (y - XB)$$

$\rightarrow$ We can calculate both of these b/c we have the data

$$2. \text{ Step 1 } \rightarrow \text{ Draw } (\sigma^2)^{(m)} \text{ from inv-gaussian } \left(\frac{n-k-1}{2}, \frac{SS\bar{E}}{2} \right), m = 1, 2, \dots, M \rightarrow M \text{ of samples}$$

$$3. \text{ Step 2 } \rightarrow \text{ Draw } \beta^{(m)} \text{ from } \text{MVN}(B, (\sigma^2)^{(m)} (X^T X)^{-1}) \rightarrow \text{can use } \text{rmvnorm}()$$

$$\Rightarrow \text{results in a set of } (\sigma^2)^{(1)}, \dots, (\sigma^2)^{(M)}$$

$$\downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow$$

$$\beta^{(1)}, \dots, \beta^{(M)}$$

- Gibbs sampler?

$$\begin{aligned} \rightarrow \text{Recall } p(\beta, \sigma^2 | y) &\propto p(y | \beta, \sigma^2) \pi(\beta, \sigma^2) \\ &= (\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} (y - X\beta)^T (y - X\beta) \right\} \times (\sigma^2)^{-1} \\ &= (\sigma^2)^{-\frac{n}{2}-1} \underbrace{\exp \left\{ -\frac{1}{2\sigma^2} (y - X\beta)^T (y - X\beta) \right\}}_{\downarrow} \end{aligned}$$

→ Need both conditional posteriors:

$$\rightarrow p(\beta | \sigma^2, y) \sim \text{MVN}(\beta, \sigma^2 (X^T X)^{-1}) \quad \text{(shown above)}$$

$$\begin{aligned} \rightarrow p(\sigma^2 | \beta, y) &\propto p(\beta, \sigma^2 | y) \\ &= (\sigma^2)^{-\frac{n}{2}-1} \exp \left\{ -\frac{1}{2\sigma^2} (y - X\beta)^T (y - X\beta) \right\} \\ &\quad \downarrow \\ &= \exp \left\{ -\frac{1}{\sigma^2} \left(\frac{(y - X\beta)^T (y - X\beta)}{2} \right) \right\} \\ &\sim \text{Inv-Gamma} \left(\frac{n}{2}, \frac{(y - X\beta)^T (y - X\beta)}{2} \right) \\ &\quad \hookrightarrow \text{rate parameter (in R)} \\ &\quad \quad \text{(not scale)} \end{aligned}$$

→ Sampling → Given the current step $(\sigma^2)^{(m)} \propto \beta^{(m)}$

→ Step 1 → Draw $(\sigma^2)^{(m+1)} \sim \text{inv-gamma} \left(\frac{n}{2}, \frac{(y - X\beta^{(m)})^T (y - X\beta^{(m)})}{2} \right)$

→ Step 2 → Draw $\beta^{(m+1)} \sim \text{MVN}(\beta, (\sigma^2)^{(m+1)} (X^T X)^{-1})$

- Interesting facts:

1. In frequentist linear regression

$$B = (X^T X)^{-1} X^T y | \beta, \sigma^2 \sim \text{MVN}(\beta, \sigma^2 (X^T X)^{-1})$$

In Bayesian analysis with noninformative priors,

$$\beta | \sigma^2, y \sim \text{MVN}(B, \sigma^2 (X^T X)^{-1})$$

2. In frequentist approach

$$\frac{B_j - \beta_j}{\sqrt{\hat{\sigma}^2 (X^T X)^{-1}_{[j+1, j+1]}}} | \beta_j, \sigma^2 \sim t_{n-k-1}$$

In Bayesian approach

$$\frac{\beta_j - B_j}{\sqrt{\hat{\sigma}^2 (X^T X)^{-1}_{[j+1, j+1]}}} | y \sim t_{n-k-1}$$

where $\hat{\sigma}^2 = \text{MSE} = y^T (I - H) y / (n - k - 1)$

Normal Models $\rightarrow Y_1, \dots, Y_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2) \Rightarrow$ same setup as for a t-test on a mean!
 $\Rightarrow$ want $\sigma^2 | Y + \mu | \sigma^2, Y$

- A special case of the linear regression model
 The normal models with unknown μ and σ^2

$$Y_i = \mu + \epsilon_i$$

$\rightarrow$ special model w/ no predictors

$\rightarrow$ can write in terms of regression as well

$$\rightarrow \text{let } \beta = \mu, \underline{y} = (Y_1, \dots, Y_n)^T, \underline{x} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}, \epsilon = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$$\Rightarrow \underline{y} = \underline{x}\beta + \epsilon$$

$\rightarrow$ thus, using previous results

$$\rightarrow \underline{B} = (\underline{x}^T \underline{x})^{-1} \underline{x}^T \underline{y} \quad \langle \underline{x}^T \underline{x} = [1, \dots, 1] \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = n \rangle$$

$$\begin{aligned} &= \frac{1}{n} \downarrow \\ &= \frac{1}{n} [1 \dots 1] \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} \\ &= \frac{1}{n} \sum_{i=1}^n Y_i \\ &\downarrow \\ &= \bar{Y} \end{aligned}$$

$$\rightarrow \sigma^2 (\underline{x}^T \underline{x})^{-1} = \frac{\sigma^2}{n}$$

$$\Rightarrow p(\epsilon | \underline{y}) \sim \mathcal{N}(\bar{Y}, \frac{\sigma^2}{n})$$

$$\rightarrow k=0 \text{ predictors} \Rightarrow \frac{n-k-1}{n} = \frac{n-1}{n}$$

$$SSE = (\underline{y} - \underline{x}\hat{\beta})^T (\underline{y} - \underline{x}\hat{\beta})$$

$$= \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

$$= \sum_{i=1}^n (Y_i - \bar{Y})^2$$

$$= (n-1) s^2 \quad \langle s^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2 \rangle$$

$$\Rightarrow \sigma^2 | \underline{y} \sim \text{Inverse-Gamma} \left(\frac{n-1}{2}, \frac{(n-1) s^2}{2} \right)$$

- Predictions

$\rightarrow$ a particular set of $\underline{x}$ values

- Suppose we want to predict for $\underline{X}_h = (1, X_{h1}, X_{h2}, \dots, X_{hk})^T, Y_h | \beta, \sigma^2 \sim \mathcal{N}(\underline{X}_h^T \beta, \sigma^2)$.
- Add a new step to original steps

$\rightarrow$ we have posterior samples + want samples from posterior predictive

$$(\sigma^2)^{(1)} + \beta^{(1)} \rightarrow \tilde{y}^{(1)} \sim \mathcal{N}(\underline{X}_h^T \beta^{(1)}, (\sigma^2)^{(1)})$$

$\vdots$

$$(\sigma^2)^{(m)} + \beta^{(m)} \rightarrow \tilde{y}^{(m)} \sim \mathcal{N}(\underline{X}_h^T \beta^{(m)}, (\sigma^2)^{(m)})$$

4.1.3 Bayesian linear regression with conjugate priors

$$\sigma^2 \sim \text{Inverse Gamma}(\alpha, \gamma)$$

$$\beta \sim \text{MVN}(\beta_0, \Sigma_0)$$

- Example: fixed values of α, γ , and Σ_0 . $\beta \sim \text{MVN}(\mathbf{0}, 100 \mathbf{I}_{(k+1) \times (k+1)})$

$$\rightarrow \underline{\beta} = (\beta_0, \beta_1, \dots, \beta_k)^T$$

$$\Rightarrow \beta_i \sim \mathcal{N}(0, 100)$$

$\rightarrow$ This structure of Σ implies no covariances

$$\rightarrow \begin{bmatrix} 100 & 0 & \dots & 0 \\ 0 & 100 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 100 \end{bmatrix} \quad \text{Diagonal} = \text{variances}$$

Secher $\hookrightarrow$ Covariances